Copper

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Cu

Found in minerals such as malachite, the color of pure copper makes it easily recognizable among metals. This element is very malleable and a good conductor of electricity. For this reason, it is useful in roofing, water pipes, coins, and electric motors.

Pellets of pure copper refined in a laboratory

Green lady

Verdigris, a mixture of several copper compounds, is the green coating that forms on copper when it is exposed to air. The Statue of Liberty, in New York City, is now green because of verdigris on its copper layer.

Verdigris-coated copper layer sits on an iron framework.

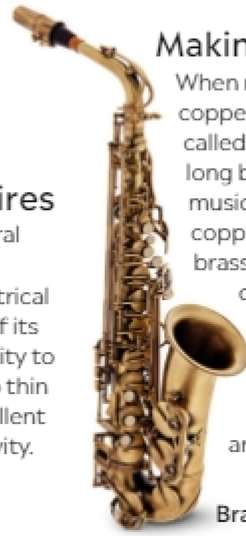
Electrical wires

Copper is a natural choice for use in circuits and electrical wiring because of its ductility (the ability to be drawn out into thin wires) and its excellent electrical conductivity.

Copper wire

Making music

When mixed with zinc, copper forms an alloy called brass, which has long been used for making musical instruments. As copper is easy to shape, brass can be formed into complicated shapes to make these instruments—they also have a good acoustic quality and are very durable.



Brass saxophone

Zinc

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Zn

A relatively common metal, zinc can be obtained from minerals such as sphalerite and smithsonite. This element is useful in many ways, mainly in compound form. Apart from being present in the alloy brass, its major use is in preventing the corrosion of steel. Its compound zinc oxide is commonly used as a white sunscreen cream.

Laboratory sample of pure zinc

In our diet

Zinc is essential in our diet. We consume it in foods such as cheese and sunflower seeds. Red meat—in the form of beef, liver, and lamb—is a particularly good source of zinc, as are herrings and oysters.

Sunflower seeds

This garden bucket's steel surface is galvanized to keep it safe from corrosion.

Galvanized bucket

Protecting steel

Steel can be shielded from corrosion by coating it with zinc—a process known as galvanization. The outer layer of zinc forms a barrier around the steel underneath, protecting it from exposure to air and water. Zinc is also a "sacrificial" metal, meaning it reacts in preference to the metal underneath it.